GB Constraint Analytics Platform

Constraint Behaviour and Storage Feasibility in the Great Britain Electricity System: A North-West Transmission Constraint Case Study

Project Findings Report (1)

Executive Summary

Objective

This study investigates whether transmission constraint behaviour creates a material operational challenge for Battery Energy Storage Systems (BESS). The project combines a GB-wide constraint screening framework with a detailed case study of selected high-burden North-West transmission constraint boundaries.

The analysis combines transmission constraint data, Balancing Mechanism data and derived operational metrics to characterise:

1. Constraint persistence
2. Operational burden
3. Constraint clustering
4. Storage survivability
5. State-of-charge dynamics

The central research question is:

> Do clustered transmission constraint episodes create a material recharge limitation for representative storage assets? The analysis was conducted in two stages. First, GB-wide constraint behaviour was characterised to identify burden concentration, persistence and operational archetypes. Second, selected North-West constraint groups were examined in detail to evaluate storage survivability, state-of-charge dynamics and recharge behaviour.

Principal Findings

Transmission constraint behaviour exhibits persistence, burden concentration and temporal clustering. Constraint events therefore cannot be treated as independent observations when assessing storage requirements.

For the analysed North-West constraint groups, representative battery simulations show that asset performance is primarily determined by power capability and energy capacity during operational episodes.

Despite meaningful clustering, historical recovery windows are generally sufficient for batteries to fully recharge between episodes. Across the analysed North-West episode population, recharge risk was not found to be a material operational limitation.

The dominant challenge for flexibility assets is therefore delivering sufficient power and energy during operational episodes rather than recovering between episodes.

Key Conclusions

1. Constraint events exhibit meaningful persistence and clustering.
2. Operational burden is highly concentrated across a small number of locations and episodes.
3. Power capability acts as the primary design constraint for representative storage assets.
4. Energy capacity becomes relevant only after power requirements have been satisfied.
5. Recharge risk was not observed to be a binding operational limitation.
6. Significant operational burden remains uncaptured even for large representative assets.

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1. Introduction

Motivation

Transmission constraints are an increasingly important feature of the Great Britain electricity system.

Battery Energy Storage Systems (BESS) are frequently proposed as a solution to congestion management, renewable integration and system flexibility challenges. However, relatively little work has been undertaken to characterise the physical structure of transmission constraint behaviour before evaluating storage feasibility.

This study addresses that gap by examining how historical constraint events evolve through time and how representative storage assets perform under observed operating conditions.

The project originated from the observation that North-West transmission constraints appeared to exhibit persistent and operationally significant behaviour that warranted further investigation. To provide system-wide context and establish a reusable analytical framework, the full GB constraint dataset was analysed. Detailed storage-feasibility and recharge analysis was then performed on selected North-West transmission constraint groups.

Research Questions

1. How persistent are transmission constraint events?
2. How severe are operational episodes?
3. Do constraint events occur independently or in clusters?
4. What power and energy capabilities are required to survive observed episodes?
5. For selected North-West constraint groups, does recharge risk materially constrain storage operation?

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2. Data and Methodology

Data Sources

The analysis integrates:

- Transmission constraint datasets
- Balancing Mechanism data
- Market signal datasets
- Derived operational metrics

These datasets were assembled within the GB Constraint Analytics Platform.

Key Metrics

Loading Ratio

A dimensionless measure of operational loading relative to observed constraint conditions.

Burden Proxy

A composite measure combining loading intensity and event duration.

Operational Episode

A cluster of constraint events separated by less than a specified recovery-gap threshold.

Survivability

An asset survives an operational episode when both:

- Power requirements remain within asset capability.
- Energy requirements remain within available capacity.

Important Interpretation Note

Power and energy values throughout this study are expressed in relative loading-ratio units and loading-hours rather than absolute MW and MWh.

Results therefore describe comparative capability requirements and relative performance rather than specific storage specifications.

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3. Constraint Behaviour

Finding 1 — Constraint Events Exhibit Persistence

Historical constraint events cannot be characterised solely by average duration.

Most events are relatively short, but a persistent tail of longer-duration events creates recurring operational challenges.

Implication

Constraint behaviour contains meaningful persistence and cannot be treated as a collection of isolated observations.

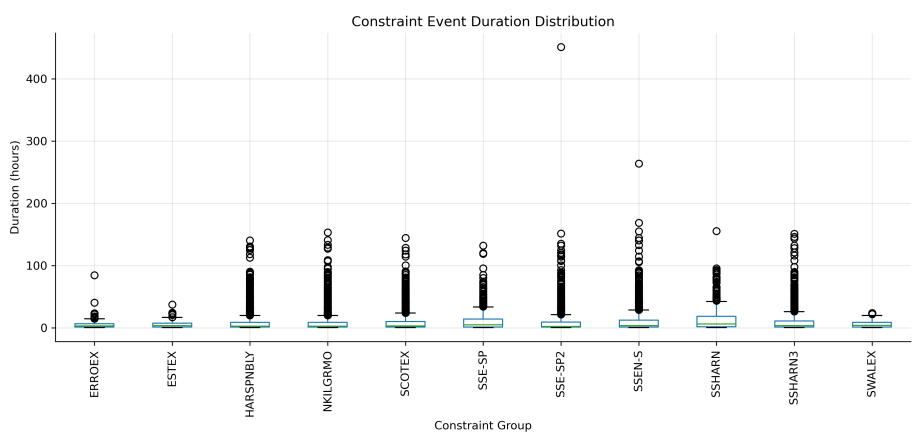

Figure 1: Figure 1 – Constraint Event Duration Distribution

Constraints are not isolated events.

Finding 2 — Operational Burden is Highly Concentrated

Operational burden is not evenly distributed across the network.

A relatively small number of transmission boundaries account for a disproportionate share of total burden. Several of the highest-burden groups were located within the North-West region and became the focus of the subsequent storage-feasibility assessment.

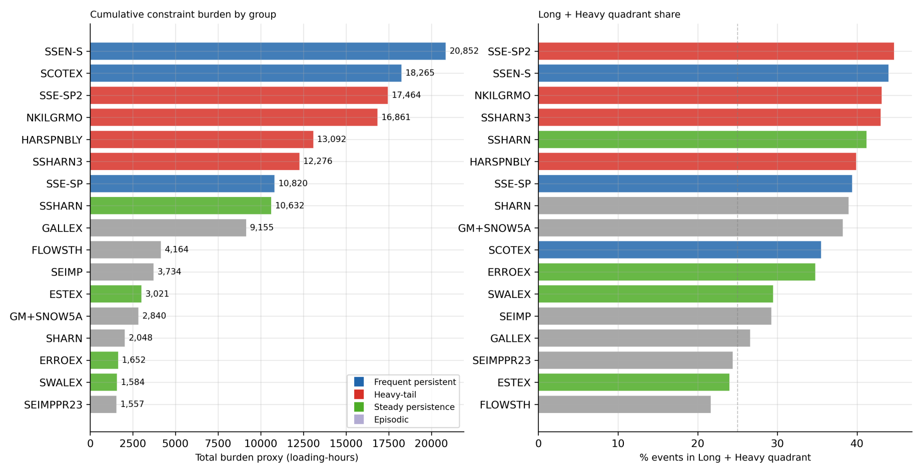

Figure 2: Figure 2 – Operational Burden Concentration

Most burden lives in a small number of locations.

Implication

Constraint management challenges are concentrated rather than uniformly distributed.

Finding 3 — Distinct Behavioural Archetypes Exist

Transmission boundaries exhibit markedly different combinations of persistence, burden and operational intensity.

Implication

No single representative constraint profile adequately describes the observed system.

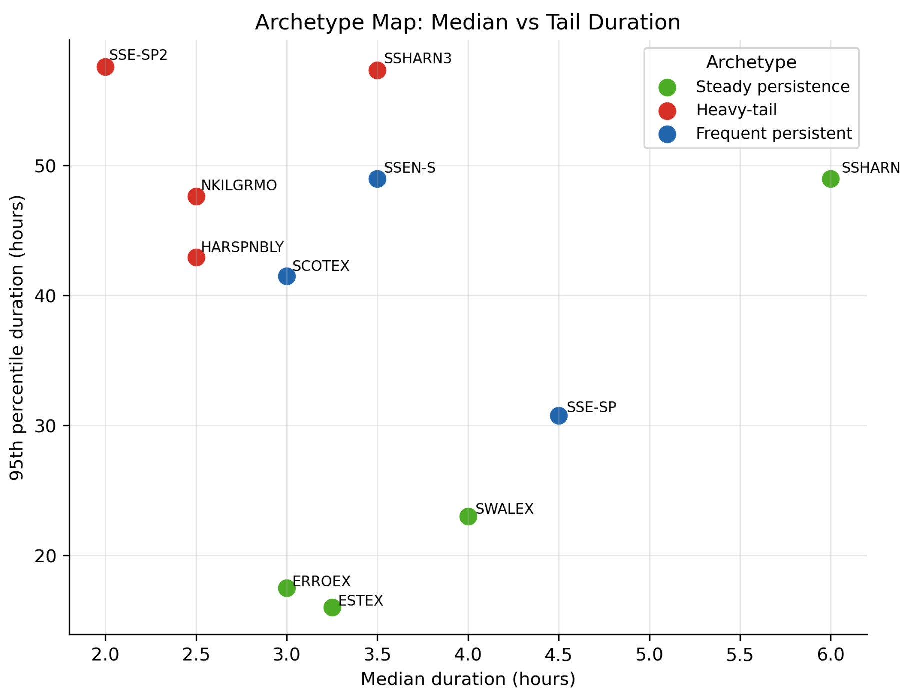

Figure 3: Figure 3 – Constraint Behaviour Archetypes

Not all boundaries behave the same way.

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4. Sequence Behaviour

Finding 4 — Constraint Events Cluster into Operational Episodes

Many apparently independent events combine into larger operational episodes when realistic recovery-gap thresholds are applied.

Implication

Constraint behaviour exhibits meaningful temporal structure.

Operational episodes provide a more realistic unit of analysis than individual settlement periods.

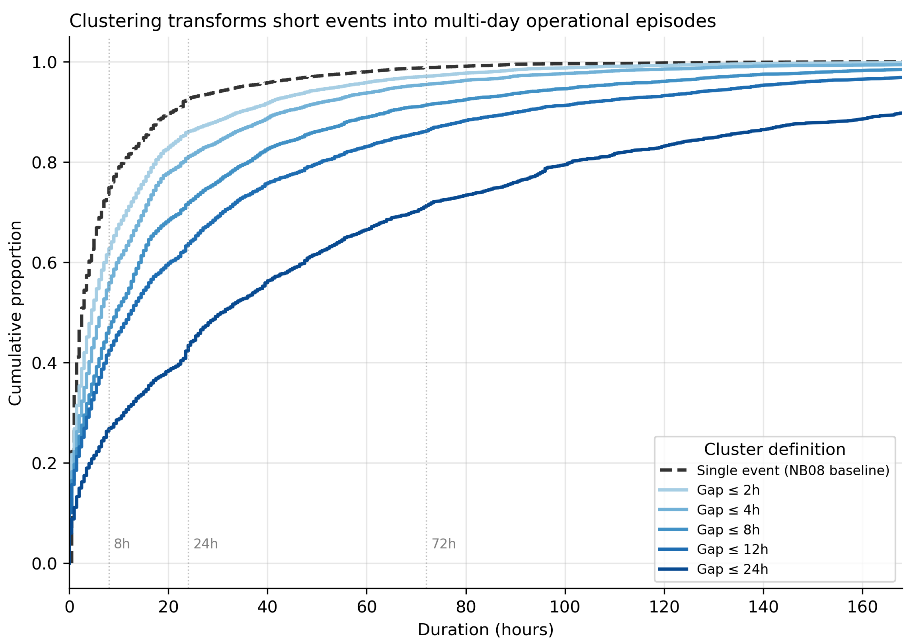

Figure 4: Figure 4 – Operational Episode Formation

Independent events become episodes.

Finding 5 — Recovery Windows Remain Substantial

Recovery periods between operational episodes are typically long relative to battery recharge requirements.

Median recovery windows are approximately one day (25.5 hrs), with upper-quartile recovery periods substantially longer.

Implication

Recharge opportunities remain available despite the presence of clustering.

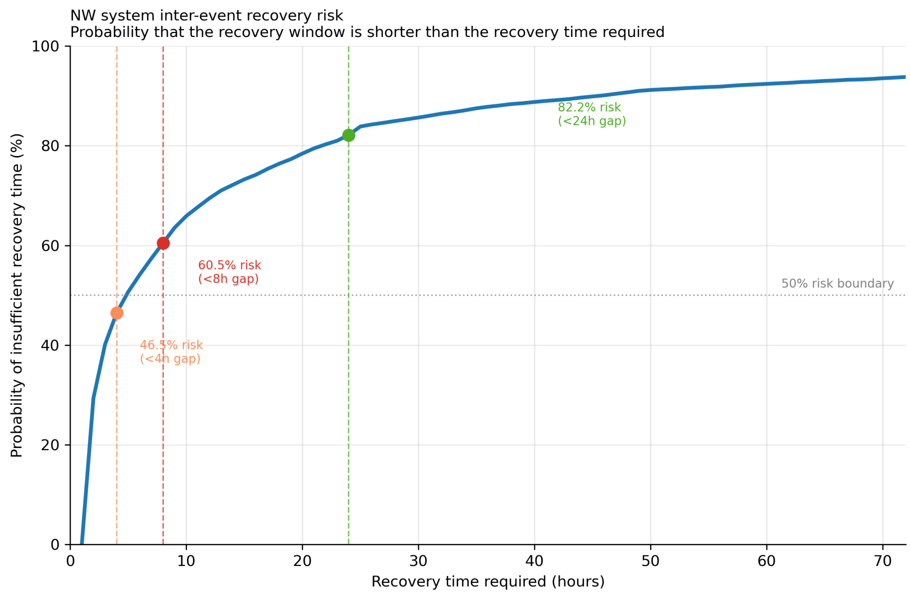

Figure 5: Figure 5 – Recovery Window Distribution

Recovery windows exist.

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5. Storage Feasibility

Finding 6 — Power Capability is the Primary Design Constraint

Historical episode outcomes indicate that power capability acts as the first feasibility hurdle.

Many smaller assets fail power-feasibility tests before energy capacity becomes relevant.

Implication

A battery may possess sufficient stored energy yet remain unable to participate because instantaneous discharge capability is inadequate.

Power capability therefore represents the primary design constraint for the analysed North-West transmission boundaries.

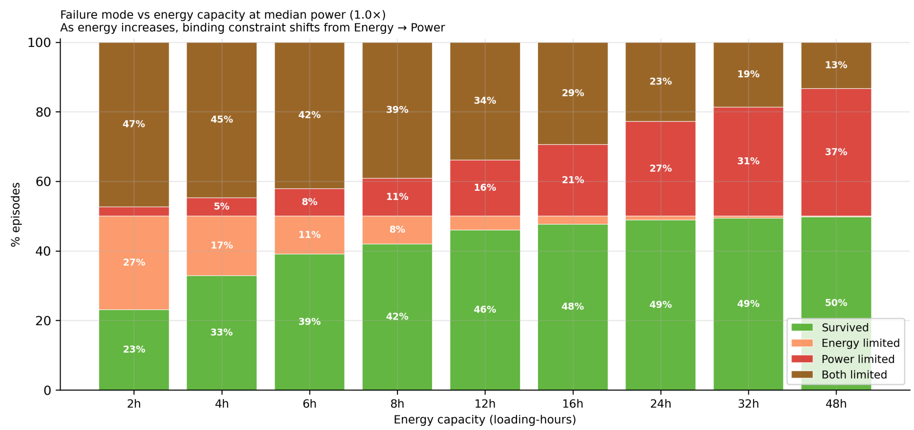

Figure 6: Figure 6 – Historical Episode Outcomes by Asset Size

Static Failure Dynamic Depletion Survived

Finding 7 — Energy Capacity Becomes the Secondary Constraint

Once power requirements are satisfied, energy limitations become increasingly important.

Larger assets experience dynamic depletion during prolonged episodes even when instantaneous power capability is sufficient.

Implication

Power and energy act as separate operational constraints.

Energy capacity becomes relevant only after power requirements have been met.

What asset sizes actually work?

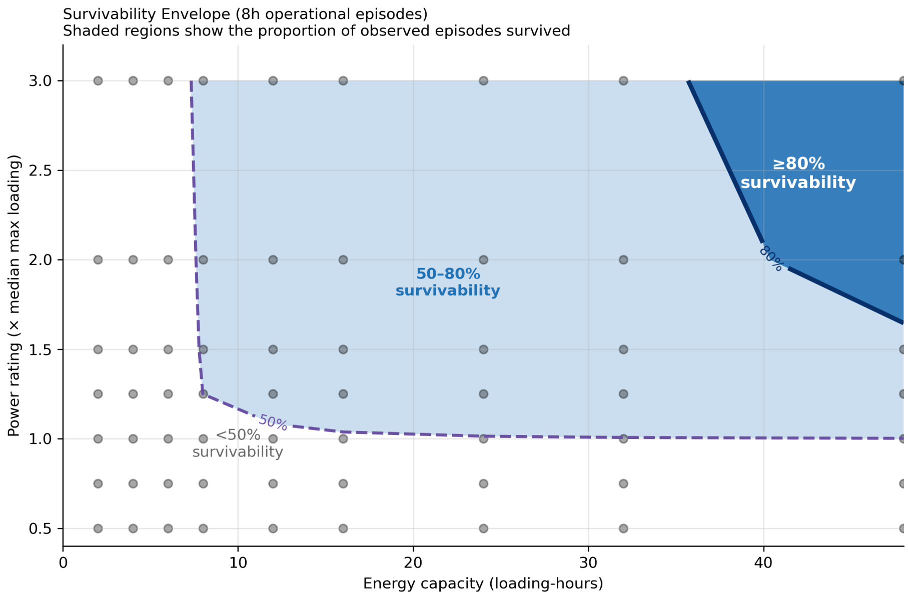

Figure 7: Figure 7 – Survivability Envelope

Finding 8 — Failure Mechanisms are Physically Distinct

Two dominant failure mechanisms emerge:

Static Failure

The asset lacks sufficient discharge capability to participate.

Dynamic Depletion

The asset begins the episode successfully but exhausts available energy before the episode concludes.

Implication

Understanding the distinction between power-limited and energy-limited behaviour is essential when designing storage assets for congestion management applications.

How recharge works.

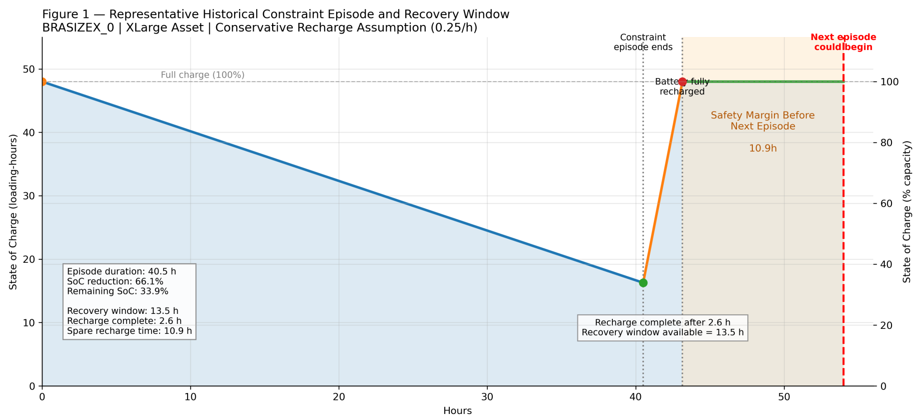

Figure 8: Figure 8 – Recharge Mechanism

6. State-of-Charge Dynamics

Finding 9 — Recharge Risk is Not a Material Constraint

State-of-charge simulations indicate that representative assets begin virtually all historical episodes fully charged.

Historical recovery windows consistently exceed recharge requirements.

Implication

Recharge limitations do not materially affect survivability across the analysed North-West constraint episodes.

Recharge works historically.

Finding 10 — Safety Margins Remain Positive

Historical safety-margin analysis demonstrates that observed recovery windows exceed required recharge durations across the analysed episode population.

No historical episode entered the recharge-risk region.

Implication

Recharge risk does not appear to be a binding operational limitation for the analysed North-West constraint groups under historical operating conditions.

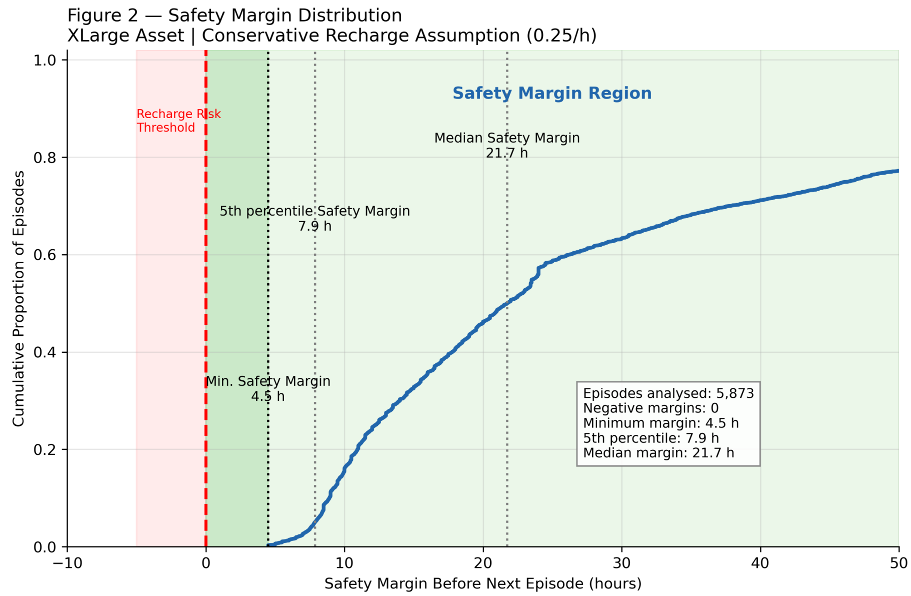

Figure 9: Figure 9 – Safety Margin Distribution

7. Discussion

Initial Hypothesis

The motivating hypothesis was:

Constraint Clustering → Recharge Pressure → Battery Limitation

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Observed Behaviour

The observed behaviour is:

Constraint Clustering → Recovery Pressure Exists → Recovery Windows Remain Sufficient → Power and Energy Requirements Dominate Outcomes

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Interpretation

The analysis demonstrates that transmission constraints create meaningful operational challenges for storage assets.

However, those challenges arise primarily from the power and energy requirements of individual operational episodes rather than from recharge limitations between episodes.

The evidence therefore does not support the hypothesis that recharge risk is a primary operational constraint for the analysed North-West transmission constraint groups under historical operating conditions.

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8. Limitations

Relative Asset Sizing

Results are expressed in loading-ratio units and loading-hours rather than absolute MW and MWh.

Simplified Intra-Episode Dynamics

State-of-charge modelling assumes simplified discharge behaviour within operational episodes.

Historical Scope

Results describe historical operating conditions and should not be interpreted as forecasts. Future system changes (higher renewable penetration, EV demand, grid outages) could alter recharge opportunity distribution

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Regional Scope

Detailed storage-feasibility findings are derived from selected North-West transmission constraint groups identified through the GB-wide screening process. Results should not automatically be assumed to apply to all transmission regions without further analysis.

Economic Value

Market value, revenue capture and flexibility economics are intentionally outside the scope of this phase of the project.

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9. Conclusions

Principal Conclusions

1. Constraint events exhibit persistence, burden accumulation and clustering.
2. Operational burden is highly concentrated across a relatively small number of locations and episodes.
3. For the analysed North-West constraint groups, power capability represents the primary design constraint for representative storage assets.
4. Energy capacity becomes important only after power requirements have been satisfied.
5. Recharge risk was not observed as a binding operational limitation across the analysed North-West episode population.
6. Constraint episodes behave as effectively independent challenges from a state-of-charge perspective.
7. Delivering sufficient power and energy during operational episodes is a more significant challenge than recovering between episodes.

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10. Future Work

Part III — Constraint Economics and Flexibility Value

This study establishes the physical operating characteristics of observed transmission constraint episodes.

Future work will investigate:

- Market conditions during operational episodes
- Constraint pricing dynamics
- Flexibility value
- Revenue capture
- Asset opportunity
- Economic significance of constraint behaviour

The engineering analysis presented here provides the physical foundation for that future economic assessment.

Appendix

Appendix Figure A1

Why power matters before energy.

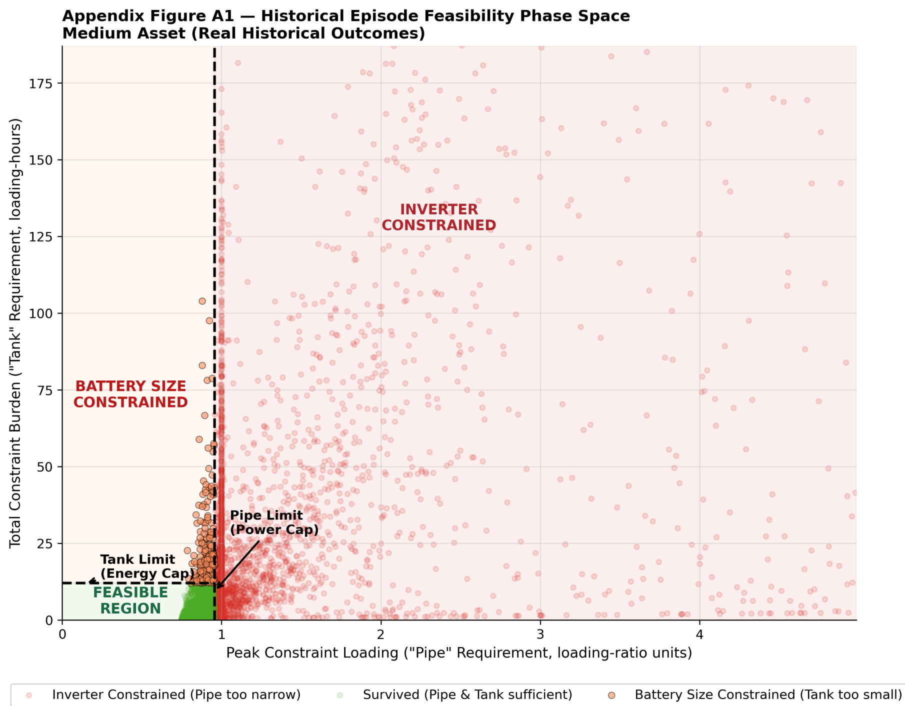

Figure 10: Appendix Figure A1 – Historical Episode Feasibility Phase Space